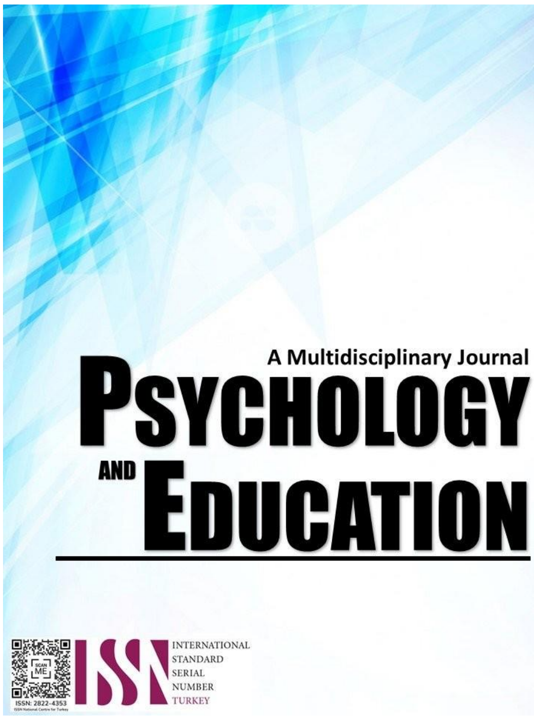
EFFECTIVENESS OF THE UTILIZATION OF INFORMATION AND COMMUNICATION TECHNOLOGY IN RELATION TO THE PUPILS' ACADEMIC PERFORMANCE: BASIS FOR ENHANCEMENT PROGRAM



PSYCHOLOGY AND EDUCATION: A MULTIDISCIPLINARY JOURNAL

Volume: 14
Pages: 715-734
Document ID: 2023PEMJ1289
DOI: 10.5281/zenodo.10032222
Manuscript Accepted: 2023-20-10

Psych Educ, 2023, **14**: 715-734, Document ID:2023 PEMJ1289, doi:10.5281/zenodo.10032222, ISSN 2822-4353
Research Article

Effectiveness of the Utilization of Information and Communication Technology in Relation to the Pupils' Academic Performance: Basis for Enhancement Program

Athena Lourdes S. Esmaquilan*, Josyl S. Edaño, Josielee Grace S. Vasquez
For affiliations and correspondence, see the last page.

Abstract

This study focus to determine the effectiveness of utilization of information and communication technology concerning the pupils' academic performance in Binalbagan District II. A descriptive research design was used in this study. The respondents were the 60 teachers in the 14 public elementary schools in Binalbagan District II. A researcher-made questionnaire was used in this study. The frequency and percentage, the mean, Mann Whitney U Test, and Pearson's *r* are the statistical tools used in this study. The findings revealed that out of 60 teacher respondents majority were older (39 years old and above, had longer(14 years and above) length of service, were bachelor's degree holders, and with both higher and lower income. The extent of effectiveness of the utilization of information and communication technology in planning the lesson, delivery of the lesson, and monitoring and evaluation was highly effective. The extent of effectiveness of the utilization of information and communication technology when grouped according to the aforementioned variables was highly effective. The academic performance of the pupils was satisfactory. In addition, no significant difference in the extent of effectiveness of the teacher's utilization of the information and communication technology in terms of age and length of service in the area of planning the lesson and delivery of the lesson. There was a significant difference in the extent of effectiveness of the teacher's utilization of the information and communication technology in terms of age and length of service in the area of planning the lesson and delivery of the lesson. Further, there was a significant difference in the extent of effectiveness of the teacher's utilization of the information and communication technology according to variables in the area of monitoring and assessment. Furthermore, there was no significant relationship between the extent of effectiveness of the utilization of information and communication technology and pupils' academic performance. Lastly, the lack of inadequate ICT equipment was the most common problem encountered by the teachers.

Keywords: *information, communication technology, academic performance, effectiveness*

Introduction

The role of technology in teaching and learning is rapidly becoming one of the most important and widely discussed issues in contemporary education policy (Rosen and Well, 2013). Most experts in the field of education agreed that, when properly used, information and communication technology hold great promise to improve teaching and learning in addition to shaping workforce opportunities.

Poole (2013) has indicated that technological illiteracy is now regarded as the new illiteracy. This has gingered a new and strong desire to equip schools with computer facilities and qualified personnel necessary to produce technologically proficient and efficient students in developed countries of the world.

Technology's use in education is becoming an absolutely important part of teaching-learning programs Some reasons for teachers to use technology in classroom instruction are to promote student agreement, to teach 221st-centuryskills, to stay current, to have hands-on interactive learning ,and to vary instructional methods. Likewise, it provides a significant contribution for teachers in planning the lessons, delivery of subject matter and monitoring and assessing pupils.

However, for many teachers in Binalbagan District II, the utilization of technology provides challenges to them. One of the common problems encountered by the teachers is the lack of or inadequate ICT equipment. Teachers used their personal gadgets to provide teaching since the school has inadequate ICT equipment. If these technologies are only adequate and used properly, they can be a tool for teachers as well as for students to help them improve academic outcomes.

Pupils' academic performance can be attributed to the effective utilization of ICT by the teachers. Also, the role of the instructor together with the effectiveness of the technology utilization can lead to advanced learning results.

For such reason, the researcher was motivated to conduct this study to assess the effectiveness of the utilization of information and communication technology about pupils' academic performance.

Research Questions

This study aimed to determine the effectiveness of the utilization of ICT about the pupils' academic performance: basis for enhancement program in Binalbagan District II. Specifically, it sought to answer the following questions:

1. What is the profile of the teachers according to the following variables:
1.1. age
1.2. length of service
1.3. highest educational attainment
1.4. average family monthly income?
2. What is the extent of effectiveness of utilization of ICT in the following areas:
2.1. planning the lesson
2.2. delivery of the lesson
2.3. monitoring and evaluation?
3. What is the extent of effectiveness of the utilization of ICT when grouped according to the aforementioned variable?
4. What is the level of academic performance of pupils?
5. Is there a significant difference on the extent of effectiveness of the teacher's utilization of ICT when grouped and compared according to the aforementioned variables?
6. Is there a significant relationship between the extent of effectiveness of utilization of ICT and the level of pupil's academic performance?
7. What are the problems encountered by the teachers?
8. Based on the findings of the study what enhancement program can be formulated?

Methodology

Research Design

The descriptive research design was used in this study in order to determine the effectiveness of the utilization of information and communication technology in relation to the pupils' academic performance in Binalbagan District II during the School Year 2017-2018. De Belen (2015) claimed that descriptive research design is appropriate for a study which aims to find out what prevail in the present conditions or relationships, held opinions and beliefs, processes and effects and developing trends. It also seeks to determine phenomena, tests hypotheses and develops generalizations, principles, or conditions and things which exist at the time of the study, it also considers past events and influences which deemed related to what is studied in the present. With nature of the present study, the descriptive research design was appropriate to be used.

Locale of the Study

This study was conducted in the 14 public elementary schools in the District of Binalbagan II namely; Biao Elementary School, Binadlan Elementary School, Bulwang Elementary School, Cabiti Elementary School, Doña Jesusa Arroyo Elementary School, Jose L. Yulo Elementary School, Mangahoycahoy Elementary School, Manolita Elementary School, Nasanagan Elementary School, Paglaum Elementary School, Payao Elementary School, Pucatod Elementary School, San Jose Elementary School and San Teodoro Elementary School.

Respondents of the Study

The respondents of the study were the 60 elementary teachers in the 14 public schools in the District of Binalbagan II, Negros Occidental.

Table 1. *Distribution of the Respondents*

Name of School	*Frequency*	*Percentage*
Biao Elementary School	5	8.3%
Binadlan Elementary School	2	3.3%
Bulwang Elementary School	2	3.3%
Cabiti Elementary School	3	5.0%
Doña Jesusa Arroyo Elementary School	5	8.3%
Jose L. Yulo Elementary School	4	6.7%
Mangahoycahoy Elementary School	3	5.0%
Manolita Elementary School	4	6.7%
Nasanagan Elementary School	3	5.0%
Paglaum Elementary School	7	11.7%
Payao Elementary School	12	20.0%
Pucatod Elementary School	3	5.0%
San Jose Elementary School	4	6.7%
San Teodoro Elementary School	3	5.0%
Total	60	100.0%

Data Gathering Instrument

The researcher of this study utilized a self- made questionnaire. Part 1 of the questionnaire was used to determine the respondent's socio demographic profile such as age, length of service, highest educational attainment and average family monthly income.

The second part was used to gather data on the extent of effectiveness of utilization of information and communication technology in terms of planning the lesson, delivery of the lesson and monitoring and evaluation.

Validity

A research instrument is regarded to be valid if it serves the purpose for which it was designed (Glenogo, 2013). The content validity involves essentially the systematic examination of the test items and its content to determine whether it covers a representative sample of the behavior domain to be measured.

The questionnaires ware further subjected to content validity analysis among the 3 panel validators. The validators were the Payao Elementary school principal, the Information and communication technology coordinator and the District supervisor of Binalbagan District II. With the reasoned judgment of these professionals, the items of the test and the entire data - gathering tool itself was rated as valid, appropriate, adequate and suited to the purpose for which it was constructed. Using the criteria set forth by Good and Scates the research instrument yielded a validity score of 5, interpreted as “Excellent”.

Reliability

Reliability is the ability to make inference from a sample to the population (Sasuman, 2015). In addition, De Belen (2015) clarifies the meaning of reliability as the degree of consistency or stability of a research instrument to produce or yield the same or identical results. The reliability of the research instrument was established through pilot testing technique in Binalbagan Elementary School utilizing 30 teachers as respondents. Using Cronbach’s Alpha, it yielded an alpha score of 0.926, interpreted as highly reliable.

Data Gathering Procedure

The researcher asked permission from the Schools Division Superintendent, Dean of the Graduate School and District Supervisor of Binalbagan District II to conduct the investigation in the concerned schools.

After the validity and reliability of the instrument was established, sufficient copies of questionnaires were reproduced and distributed to the respondents of the study. During the actual conduct of the study, the researcher personally administered the questionnaires and sees to it that it was completely filled out. After the conduct, the researcher tabulated the responses of the respondents and submit it to the statistician for statistical treatment.

Analytical Scheme

The following analytical schemes were used:

For Problem 1 which aimed to determine the profile of the teachers according to variables such as age, length of service, highest educational attainment and average family monthly income, the descriptive analytical scheme was applied.
For Problem 2, which aimed to determine the extent of effectiveness of the utilization of information and communication technology in terms of planning the lesson, delivery of the lesson and monitoring and evaluation, the descriptive analytical scheme was applied.
For Problem 3, which aimed to determine the extent of effectiveness of the utilization of ICT when grouped according to the aforementioned variables, the descriptive analytical scheme was used.
For Problem 4, which aimed to determine the academic performance of the pupils, the descriptive analytical scheme was applied.
For Problem 5 which sought to determine if there is a significant difference on the extent of effectiveness of utilization of ICT when grouped and compared according to the aforementioned variables, the comparative analytical scheme was used.
For Problem 6 which sought to determine if there is a significant relationship between the extent of effectiveness of utilization of ICT and the level of pupils’ academic performance, the relational analytical scheme was used.

Results and Discussion

Profile of the Respondents

A total of 60 teachers were surveyed to determine the effectiveness on the utilization of information and communication technology. When grouped according to age, 27 or 45 percent were younger (38 years old and below and 33 or 55 percent were older (39 years old and above). In terms of length of service 27 or 45 percent had a shorter (13 years and below) length of service and 33 or 55 percent has a longer (14 years and above) length of service. In terms of highest educational attainment, 35 or 58.3 percent had bachelor degree and 25 or 41.7 percent had master’s degree. Lastly, in terms of average family monthly income, 30 or 50 percent had a lower (19,000 and below) income and 30 or 50 percent had a higher (19,001 and above) income.

The result implies that majority of the teachers were older, had a shorter length of service, had bachelor’s degree and belongs to higher and lower income group.

Table 2. *Profile of the Respondents*

Variable	*Category*	*Frequency*	*Percentage*
Age	younger (38 years old and below)	27	45.0
	older (39 years old and above)	33	55.0
	Total	60	100.0
Length of Service	shorter (13 years and below)	27	45.0
	longer (14 years and above)	33	55.0
	Total	60	100.0
Highest Educational Attainment	Bachelor	35	58.3
	Masters	25	41.7
	Total	60	100.0
Average Family Monthly Income	lower (19,000 and below)	30	50.0
	higher (19,001 and above)	30	50.0
	Total	60	100.0

Extent of Effectiveness of the Utilization of Information and Communication Technology

The extent of effectiveness of the utilization of information and communication technology was evaluated in terms of planning the lesson. Table 3 presents the statistical results. The highest mean score obtained by the respondents was 4.60, interpreted as "Highly Effective" was on Item No. 10, "Provides interactive learning experiences". On the contrary, the lowest mean score was 4.35, interpreted as "Highly Effective" was on Item No. 6, "Automates a lot of teachers and students' tedious tasks".

The result implies that the utilization of ICT by the teachers allows students to experience interactive learning and makes teachers and students task done easily. The overall mean score was 4.47, interpreted as "Highly Effective". The result denotes that the extent of effectiveness of the utilization of information and communication technology in the area of lesson planning was "High".

The result was supported by the statement of (Ofsted, 2015) that ICT allows for a higher quality lessons through collaboration with teachers in planning and preparing resources. Further, Pedretti (2015) proves that teachers who used educational technology felt more successful in the delivery of learning and motivated the pupils to learn more and have increased self- confidence and self-esteem. It is also confirmed that many students found learning in a technology-enhanced setting more stimulating and much better than in a traditional classroom environment.

Table 3. *Extent of Effectiveness of the Utilization of Information and Communication Technology in terms of Planning the Lesson*

Items	*Mean*	*Interpretation*
1. Improves teaching with more updated materials.	4.47	Highly Effective
2. Prepares of teaching resources and materials.	4.47	Highly Effective
3. Develops practical skills, including creating presentations, learning to differentiate reliable from unreliable sources on the Internet, maintaining proper online etiquette, and writing emails.	4.45	Highly Effective
4. Enhances the traditional ways of teaching and to keep students more engaged.	4.52	Highly Effective
5. Provides countless resources for enhancing education and making learning more fun and effective.	4.47	Highly Effective
6. Automates a lot of teachers / students tedious tasks.	4.35	Highly Effective
7. Instant access to information that can supplement teaching and learning experience.	4.43	Highly Effective
8. Transforms the present isolated, teacher-centered and text-bound classrooms into rich, student-focused, interactive knowledge environments.	4.48	Highly Effective
9. Provides students with a nearly endless supply of information and resources.	4.47	Highly Effective
10. Provides interactive learning experiences.	4.60	Highly Effective
Over-all Mean	4.47	Highly Effective

Extent of Effectiveness of the Utilization of Information and Communication Technology in Terms of Delivery of Lesson

Table 4 shows the extent of effectiveness of the utilization of information and communication technology in terms of delivery of lesson. The result showed that ICT encourages a more active participation in the learning process which can be hard to achieve through a traditional lecture environment, as evidence by the mean score of 4.67, interpreted as "Highly Effective". Meanwhile, the teachers rated the utilization of ICT as "Highly Effective in facilitating comprehension of abstract concept and life skills. The mean was 4.35 evaluated as the lowest score yielded among the 10 statements used as criteria in evaluating the extent of effectiveness of the utilization of information and communication technology in terms of delivery of lesson.

Table 4. *Extent of Effectiveness of the Utilization of Information and Communication Technology in Terms of Delivery of Lesson*

Items	*Mean*	*Interpretation*
1. Improves the quality of teaching.	4.57	Highly Effective
2. Makes delivery of lesson more interactive.	4.48	Highly Effective
3. Helps students learn at a rate that is comfortable to them, and that allows them to retain information.	4.48	Highly Effective
4. Facilitates comprehension of abstract concept and life skills.	4.35	Highly Effective
5. Enables students' to express their ideas and thoughts better.	4.50	Highly Effective
6. Promotes active and engaging lesson for students' best learning experience.	4.47	Highly Effective
7. Enables the students' to be more active and engaged in the lesson.	4.45	Highly Effective
8. Improves motivation and engagement in the learning process.	4.48	Highly Effective
9. Provides different opportunities to make learning more fun and enjoyable in terms of teaching same things in new ways.	4.52	Highly Effective
10. Encourages a more active participation in the learning process which can be hard to achieve through a traditional lecture environment.	4.67	Highly Effective
Over-all Mean	4.50	Highly Effective

The result implies that the utilization of ICT in the delivery of lessons improves comprehension of the students and enables the students to appreciate the lesson that the traditional way of handling it. Over-all mean was 4.50, interpreted as "Highly Effective". The result indicates that the extent of effectiveness of the utilization of information and communication technology in terms of delivery of lesson was "High".

According to Maron (2014), ICT gives educators the opportunity to transform the way learning happens, and enable student development. ICT presents a range of tools that teachers or lecturers use to present and display as part of their teaching and help educators interact with students as well as engage them in a more meaningful way. These technological tools can be purposefully designed for education, for example, software or hardware used in the context such as word processors and spreadsheets.

Extent of Effectiveness of the Utilization of Information and Communication Technology in terms of Monitoring and Assessment

One of the objectives of this study was to determine the extent of effectiveness of the utilization of Information and Communication Technology in terms of monitoring and assessment. The mean scores with their verbal interpretations are presented in Table 5.

When teachers gauge the extent of effectiveness of the utilization of Information and Communication Technology in terms of monitoring and assessment, they obtained the highest mean score of 4.67, interpreted as "Highly Effective" on Items Nos. 1 and 9, "Automates computation of students' grades" and "Provides of appropriate record-keeping and thereby monitors students' progress". However, they obtained the lowest mean score of 4.53, interpreted likewise as "Highly Effective" on Item No. 6, "Designs testing materials". The overall mean score was 4.62, interpreted as "Highly Effective" which denoted that the extent of effectiveness of the utilization of ICT in terms of Monitoring and Assessment was "High". The result implies that the utilization of ICT by the teachers is "Highly Effective" in Monitoring and Assessment.

Flyn (2016) contented that once teachers have brought the technicalities involved with classroom- related hardware and software resources, they can enhance their teaching. This can be used to create additional teaching resources.

Martin (2015) construed that the use of ICT for assessment purposes can also release valuable teacher time. Within schools, those specific technological developments such as connectivity via broadband access to internet will regard to the personalization of the learning experience. A key strand to embed ICT in schools has been and that of networking within schools and also across the education sector.

Table 5. *Extent of Effectiveness of the Utilization of Information and Communication Technology in Terms of Monitoring and Assessment*

Items	Mean	Interpretation
1. Automates computation of students' grades.	4.67	Highly Effective
2. Grading software and online assessments can help teachers save a lot time.	4.65	Highly Effective
3. Encodes of formative and summative assessment.	4.65	Highly Effective
4. Stores and recording information about students' performance.	4.63	Highly Effective
5. Encodes of test questions and reproduction.	4.57	Highly Effective
6. Designs testing materials.	4.53	Highly Effective
7. Submits of school reports.	4.62	Highly Effective
8. Uploads students' information and grades.	4.62	Highly Effective
9. Provides of appropriate record-keeping and thereby monitors students' progress.	4.67	Highly Effective
10. Use of spreadsheets to track student work and also track their teaching plans.	4.60	Highly Effective
Over-all Mean	4.62	Highly Effective

Extent of Effectiveness of the Utilization of Information and Communication Technology in Terms of Planning the Lesson According to Age

Table 6 presents the extent of effectiveness of the utilization of information and communication technology in terms of planning the lesson according to age. Younger teachers obtain a highest mean score of 4.70, interpreted as "Highly Effective" on Item No. 10, "Provides of interactive learning experiences". On the other hand, a lowest mean score yielded was on item No. 8, Transforms the present isolated, teacher-centered and text-bound classrooms into rich, student-focused, interactive knowledge environments, interpreted as "Highly Effective".

The mean score was 4.56, interpreted as "Highly Effective", indicating that the extent of effectiveness of the utilization of information and communication technology in terms of planning the lesson according to age was "High".

When older teachers evaluated the extent of effectiveness of the utilization of information and communication technology in terms of planning the lesson, they obtained a highest mean score of 4.55, interpreted as "Highly Effective", on the contrary, a lowest mean score of 4.18 interpreted as effective was obtained in Item No. 6, "Automates a lot of teachers and students tedious tasks". The overall mean score was 4.39, interpreted as "Highly Effective". The result implies that the extent of effectiveness of the utilization of ICT in terms of planning the lesson according to age was "High".

Table 6. *Extent of Effectiveness of the Utilization of Information and Communication Technology in Terms of Planning the Lesson According to Age*

	Younger		Older	
Items	Mean	Interpretation	Mean	Interpretation
1. Improves teaching with more updated materials.	4.59	Highly Effective	4.39	Highly Effective
2. Prepares of teaching resources and materials.	4.52	Highly Effective	4.42	Highly Effective
3. Develops practical skills, including creating presentations, learning to differentiate reliable from unreliable sources on the Internet, maintaining proper online etiquette, and writing emails.	4.48	Highly Effective	4.42	Highly Effective
4. Enhances the traditional ways of teaching and to keep students more engaged.	4.63	Highly Effective	4.39	Highly Effective
5. Provides countless resources for enhancing education and making learning more fun and effective.	4.52	Highly Effective	4.39	Highly Effective
6. Automates a lot of teachers / students tedious tasks	4.56	Highly Effective	4.18	Effective
7. Instant access to information that can supplement teaching and learning experience.	4.59	Highly Effective	4.27	Highly Effective
8. Transforms the present isolated, teacher-centered and text-bound classrooms into rich, student-focused, interactive knowledge environments.	4.44	Highly Effective	4.52	Highly Effective
9. Provides students with a nearly endless supply of information and resources.	4.52	Highly Effective	4.39	Highly Effective
10. Provides of interactive learning experiences.	4.70	Highly Effective	4.55	Highly Effective
Over-all Mean	4.56	Highly Effective	4.39	Highly Effective

The result implies that teachers, regardless of age, believe that ICT has contributed much on monitoring and assessment of pupils especially in the computation of students' grades, designing of testing materials and keeping of pupils records and progress.

According to Roca (2015), from a teaching point of view, teachers used these devices to deliver to a whole class, and could use the digital content effectively that was available to them. Teachers also reported that ICT offered them enhanced resources to support learning through teaching. The levels of interaction, the immediacy and the ability to refresh work, were all indicated as ways in which ICT could enhance the range of teaching approaches taken. In some colleges, teachers were expecting more of the students used ICT- whether this was due to the higher pace in lessons, work being done more quickly.

Extent of Effectiveness of the Utilization of Information and Communication Technology in Terms of Planning the Lesson According to Length of Service

Table 7 shows the extent of effectiveness of the utilization of Information and Communication Technology in terms of planning the lesson according to length of service. Teachers who belong to the group with shorter length of service obtained the highest mean score of 4.70 in Item No. 10, "Provides interactive learning experiences". The lowest mean score of 4.44, interpreted as "Highly Effective" was in Item No. 8, "Transforms the present isolated, teacher-centered and text-bound classrooms into rich, student-focused, interactive knowledge environments".

The overall mean score was 4.56, interpreted as "Highly Effective". The result indicates that the teachers with shorter length of service perceived that the utilization of information and communication technology in terms of planning the Lesson was "Highly Effective".

When teachers with longer length of service was assessed, they obtained the highest mean score of 4.52, interpreted as "Highly Effective on both Items 8 and 10, "Transforms the present isolated, teacher-centered" and "Text-bound classrooms into rich, student-focused, interactive knowledge environments and Provides interactive learning experiences". "On the contrary, they obtained the lowest mean score of 4.18, interpreted as effective on Item No. 6, Automates a lot of teachers and students tedious tasks. The overall mean score was 4.40, interpreted as "Highly Effective".

The result denotes that teachers with longer length of service find out that the ICT is highly effective in planning the lesson . The result was supported by the statement of Lai and Pratt (2015) the most obvious effects of ICT use for teachers were not the change of teaching philosophy or pedagogy, as one might hope, but the improved efficiency of management and administration of teaching, accessing resources for preparing teaching materials and presenting lessons.

Table 7. *Extent of Effectiveness of the Utilization of Information and Communication Technology in Terms of Planning the Lesson According to Length of Service*

Items	*Shorter*		*Longer*	
	Mean	*Interpretation*	*Mean*	*Interpretation*
1. Improves teaching with more updated materials.	4.59	Highly Effective	4.36	Highly Effective
2. Prepares of teaching resources and materials.	4.52	Highly Effective	4.42	Highly Effective
3. Develops practical skills, including creating presentations, learning to differentiate reliable from unreliable sources on the Internet, maintaining proper online etiquette, and writing emails.	4.48	Highly Effective	4.42	Highly Effective
4. Enhances the traditional ways of teaching and to keep students more engaged.	4.63	Highly Effective	4.42	Highly Effective
5. Provides countless resources for enhancing education and making learning more fun and effective.	4.52	Highly Effective	4.42	Highly Effective
6. Automates a lot of teachers / students tedious tasks	4.56	Highly Effective	4.18	Effective
7. Instant access to information that can supplement teaching and learning experience.	4.59	Highly Effective	4.30	Highly Effective
8. Transforms the present isolated, teacher-centered and text-bound classrooms into rich, student-focused, interactive knowledge environments.	4.44	Highly Effective	4.52	Highly Effective
9. Provides students with a nearly endless supply of information and resources.	4.52	Highly Effective	4.42	Highly Effective
10. Provides interactive learning experiences.	4.70	Highly Effective	4.52	Highly Effective
Over all Mean	4.56	Highly Effective	4.40	Highly Effective

Extent of Effectiveness of the Utilization of Information and Communication Technology in Terms of Planning the Lesson According to Highest

Educational Attainment

Table 8 presents the extent of effectiveness of the utilization of information and communication technology in terms of planning the lesson according to highest educational attainment. Teachers with bachelor's degree obtained the highest mean score of 4.46, interpreted as "Highly Effective" on Item No. 10, "Provides interactive learning experiences". The lowest mean score obtained was 4.17, interpreted as "Effective" on Item No. 6, "Automates a lot of teachers and students tedious tasks". The overall mean score was 4.29, interpreted as "Highly Effective". The result implies that teachers with bachelor's degree viewed that the ICT is "Highly Effective" in planning the lesson.

When teachers with master's degree gauged the extent of effectiveness of the utilization of information and communication technology in terms of planning the lesson, they obtain a highest mean score of 4.88, interpreted as "Highly Effective", conversely, they obtained the lowest mean score of 4.60, interpreted as "Highly Effective" on Items 6 and 7, "Automates a lot of teachers and students tedious tasks and Instant access to information that can supplement teaching and learning experience". The result implies that teachers with master's degree commonly described that ICT is "Highly Effective" in planning the lesson.

Table 8. *Extent of Effectiveness of the Utilization of Information and Communication Technology in Terms of Planning the Lesson According to Highest Educational Attainment*

	Bachelor		*Masters*	
Items	*Mean*	*Interpretation*	*Mean*	*Interpretation*
1. Improves teaching with more updated materials.	4.31	Highly Effective	4.68	Highly Effective
2. Prepares of teaching resources and materials.	4.31	Highly Effective	4.68	Highly Effective
3. Develops practical skills, including creating presentations, learning to differentiate reliable from unreliable sources on the Internet, maintaining proper online etiquette, and writing emails.	4.26	Highly Effective	4.72	Highly Effective
4. Enhances the traditional ways of teaching and to keep students more engaged.	4.26	Highly Effective	4.88	Highly Effective
5. Provides countless resources for enhancing education and making learning more fun and effective.	4.23	Effective	4.80	Highly Effective
6. Automates a lot of teachers / students tedious tasks	4.17	Effective	4.60	Highly Effective
7. Instant access to information that can supplement teaching and learning experience.	4.31	Highly Effective	4.60	Highly Effective
8. Transforms the present isolated, teacher-centered and text-bound classrooms into rich, student-focused, interactive knowledge environments.	4.31	Highly Effective	4.72	Highly Effective
9. Provides students with a nearly endless supply of information and resources.	4.31	Highly Effective	4.68	Highly Effective
10. Provides interactive learning experiences.	4.46	Highly Effective	4.80	Highly Effective
Over-all Mean	4.29	Highly Effective	4.72	Highly Effective

Selwood and Pilkington (2015) found that teachers believed that ICT can assist in reducing their workload and make them more productive. Use of ICT as Edwards (2015) stressed well, requires a different set of management techniques. These techniques consist of ways to empower students to regulate their own classroom activities responsibly.

Extent of Effectiveness of the Utilization of Information and Communication Technology in Terms of Planning the Lesson According to Average Family Monthly Income

When grouped according to family income, teachers with lower income rated the utilization of ICT in terms of lesson planning as highly effective as evidence by the highest mean score of 4.37 on Item No. 10, "Provides interactive learning experiences". However, a lowest mean score of 4.13, interpreted as "Highly Effective" was obtained on Item No. 5 and 6, "Automates a lot of teachers and students tedious tasks" and "Provides countless resources for enhancing education and making learning more fun and effective". The overall mean score was 4.21, interpreted as "Effective".

The result implies that teachers with lower family income have lesser fund and resources for ICT which makes lesson planning more tedious for them and the pupils. Likewise, resources for enhancing education and making learning more fun and effective is limited.

Teachers with higher family monthly income obtained the highest mean score of 4.83 on Item No. 10, "Provides interactive learning experiences". Conversely, the lowest mean score of 4.57, interpreted as "Highly Effective" on Item No. 6, "Automates a lot of teachers and students tedious tasks". The overall mean was 4.73, interpreted as "Highly Effective". The result indicates that teachers with higher family monthly income find out that ICT was “Highly Effective” in planning the lesson.

Table 9. *Extent of Effectiveness of the Utilization of Information and Communication Technology in Terms of Planning the Lesson According to Average Family Monthly Income*

Items	*Mean*	*Lower Interpretation*	*Mean*	*Higher Interpretation*
1. Improves teaching with more updated materials.	4.20	Effective	4.73	Highly Effective
2. Prepares of teaching resources and materials.	4.17	Effective	4.77	Highly Effective
3. Develops practical skills, including creating presentations, learning to differentiate reliable from unreliable sources on the Internet, maintaining proper online etiquette, and writing emails.	4.20	Effective	4.70	Highly Effective
4. Enhances the traditional ways of teaching and to keep students more engaged.	4.23	Effective	4.80	Highly Effective
5. Provides countless resources for enhancing education and making learning more fun and effective.	4.13	Effective	4.80	Highly Effective
6. Automates a lot of teachers / students tedious tasks.	4.13	Effective	4.57	Highly Effective
7. Instant access to information that can supplement teaching and learning experience.	4.27	Highly Effective	4.60	Highly Effective
8. Transforms the present isolated, teacher-centered and text-bound classrooms into rich, student-focused, interactive knowledge environments.	4.20	Effective	4.77	Highly Effective
9. Provides students with a nearly endless supply of information and resources.	4.20	Effective	4.73	Highly Effective
10. Provides interactive learning experiences.	4.37	Highly Effective	4.83	Highly Effective
Over-all Mean	4.21	Effective	4.73	Highly Effective

ICTs have the potential to accelerate, enrich, and deepen skills, to motivate and engage students, to help relate school experience to work practices, create economic viability for tomorrow's workers, as well as strengthening teaching and helping schools change(Willona,2015). The study of Pelbert(2016), revealed parallel result with the present study. The result reveals that teachers with higher income proves that ICT is effective since most of them has the capacity to buy the gadgets and educational technologies.

Extent of Effectiveness of the Utilization of Information and Communication Technology in terms of Delivery of Lesson According to Age

Table 10 shows the extent of effectiveness of the utilization of information and communication technology in terms of delivery of lesson according to age. Teachers that belonged to the younger age group obtained a highest mean score of 4.78 interpreted as “Highly Effective” on Items No. 10, “Encourages a more active participation in the learning process” which can be hard to achieve through a traditional lecture environment. They garnered the lowest mean score of 4.41, interpreted as “Highly Effective” on Item No. 4, Facilitates comprehension of abstract concept and life skills. The overall mean score was 4.60, interpreted as “Highly Effective”, implying that the teachers that belonged to the younger age group believes that the ICT was highly effective in the delivery of lesson.

Older group of teachers garnered the highest mean score of 4.58, interpreted as “Highly Effective” on Items No. 10, “Encourages a more active participation in the learning process which can be hard to achieve through a traditional lecture environment”. They obtained the lowest mean score of 4.30, interpreted as “Highly Effective” on item number 4, Facilitates comprehension of abstract concept and life skills. The overall mean score was 4.42, interpreted as “Highly Effective”, which implies that the older teachers found that ICT was highly effective in the delivery of lessons. The adventure and advancement of new technologies (ICT) has challenge the traditional method and process of teaching and learning and have also change the way education is managed to a more flexible, friendly and simplified form. The study of Froelan (2015), revealed that ICT utilization had influenced teachers’ effectiveness in the delivery of lessons.

Table 10. *Extent of Effectiveness of the Utilization of Information and Communication Technology in Terms of Delivery of Lesson According to Age*

Items	Younger Mean	Younger Interpretation	Older Mean	Older Interpretation
1. Improves the quality of teaching.	4.67	Highly Effective	4.48	Highly Effective
2. Makes delivery of lesson more interactive.	4.52	Highly Effective	4.45	Highly Effective
3. Helps students learn at a rate that is comfortable to them, and that allows them to retain information.	4.48	Highly Effective	4.48	Highly Effective
4. Facilitates comprehension of abstract concept and life skills.	4.41	Highly Effective	4.30	Highly Effective
5. Enables students' to express their ideas and thoughts better.	4.63	Highly Effective	4.39	Highly Effective
6. Promotes active and engaging lesson for students' best learning experience.	4.59	Highly Effective	4.36	Highly Effective
7. Enables the students' to be more active and engaged in the lesson.	4.59	Highly Effective	4.33	Highly Effective
8. Improves motivation and engagement in the learning process.	4.67	Highly Effective	4.33	Highly Effective
9. Provides different opportunities to make learning more fun and enjoyable in terms of teaching same things in new ways.	4.63	Highly Effective	4.42	Highly Effective
10. Encourages a more active participation in the learning process which can be hard to achieve through a traditional lecture environment.	4.78	Highly Effective	4.58	Highly Effective
Over-all Mean	4.60	Highly Effective	4.42	Highly Effective

Extent of Effectiveness on the Utilization of Information and Communication Technology in Terms of Delivery of Lesson According to Length of Service

Table 11 shows the extent of effectiveness on the utilization of information and communication technology in terms of delivery of lesson according to length of service. Teachers with shorter length of service obtained the highest mean score of 4.74 on Item No. 10, "Encourages a more active participation in the learning process which can be hard to achieve through a traditional lecture environment, on the other hand", a lowest mean score of 4.37, interpreted as "Highly Effective" was on Item No. 4, "Facilitates comprehension of abstract concept and life skills".

The overall mean score was 4.56, interpreted as "Highly Effective", which denotes that teachers with shorter length of service find out that ICT was highly effective in the delivery of lesson.

Table 11. *Extent of Effectiveness of the Utilization of Information and Communication Technology in Terms of Delivery of Lesson According to Length of Service*

Items	Shorter Mean	Shorter Interpretation	Longer Mean	Longer Interpretation
1. Improves the quality of teaching.	4.63	Highly Effective	4.52	Highly Effective
2. Makes delivery of lesson more interactive.	4.48	Highly Effective	4.48	Highly Effective
3. Helps students learn at a rate that is comfortable to them, and that allows them to retain information.	4.52	Highly Effective	4.45	Highly Effective
4. Facilitates comprehension of abstract concept and life skills.	4.37	Highly Effective	4.33	Highly Effective
5. Enables students' to express their ideas and thoughts better.	4.59	Highly Effective	4.42	Highly Effective
6. Promotes active and engaging lesson for students' best learning experience.	4.56	Highly Effective	4.39	Highly Effective
7. Enables the students' to be more active and engaged in the lesson.	4.56	Highly Effective	4.36	Highly Effective
8. Improves motivation and engagement in the learning process.	4.63	Highly Effective	4.36	Highly Effective
9. Provides different opportunities to make learning more fun and enjoyable in terms of teaching same things in new ways.	4.56	Highly Effective	4.48	Highly Effective
10. Encourages a more active participation in the learning process which can be hard to achieve through a traditional lecture environment.	4.74	Highly Effective	4.61	Highly Effective
Over- all Mean	4.56	Highly Effective	4.44	Highly Effective

When teachers with longer length of service evaluated the extent of effectiveness of ICT utilization in the delivery of lesson, they garnered the highest mean score of 4.61, interpreted as "Highly Effective". Meanwhile, a lowest mean score of 4.33, interpreted as "Highly Effective" was on item No. 4, "Facilitates comprehension of abstract concept and life skills".

The overall mean score was 4.44, interpreted as "Highly Effective". The result indicates that teachers with longer length of service find that ICT is highly effective in the delivery of lessons. In the study of

Ram(2015), younger and older teachers' perception on the effectiveness of ICT in the delivery of lesson was evaluated, the result revealed that teachers with longer length of service found ICT to be effective in the delivery of lessons.

Extent of Effectiveness of the Utilization of Information and Communication Technology in terms of Delivery of Lesson According to Highest Educational Attainment

Table 12 presents the extent of effectiveness of the utilization of information and communication technology in terms of delivery of lesson according to highest educational attainment. Teachers with bachelor's degree obtained a highest mean score of 4.54, interpreted as "Highly Effective" on Item No. 10, "Encourages a more active participation in the learning process" which can be hard to achieve through a traditional lecture environment. The lowest mean score of 4.23, interpreted as "Effective" was on Item No. 4, Facilitates comprehension of abstract concept and life skills. The overall mean score was 4.35, interpreted as "Highly Effective", indicating that teachers with bachelor's degree find that ICT is "Highly Effective" in the delivery of lesson.

When teachers with master's degree evaluated the extent of effectiveness of the utilization of information and communication technology in terms of delivery of lesson, they obtained a highest mean core of 4.84, interpreted as Highly Effective. On the contrary, a lowest mean score was obtained on Item No. 4, "Facilitates comprehension of abstract concept and life skills". The result indicates that the utilization of information and communication technology by the teachers with master's degree in terms of delivery of lesson is "Highly Effective".

According to Lai and Pratt (2016), the most obvious effects of ICT use for teachers were not the change of teaching philosophy or pedagogy, as one might hope, but the improved efficiency of management and administration of teaching, accessing resources for preparing teaching materials and presenting lessons. Selwood and Pilkington (2015) found that teachers believed that ICT can assist in reducing their workload and make them more productive.

Table 12. *Extent of Effectiveness of the Utilization of Information and Communication Technology in Terms of Delivery of Lesson According to Highest Educational Attainment*

Degree	*With Bachelor's*		*With Masters'*	
Items	*Mean*	*Interpretation*	*Mean*	*Interpretation*
1. Improves the quality of teaching.	4.46	Highly Effective	4.72	Highly Effective
2. Makes delivery of lesson more interactive.	4.31	Highly Effective	4.72	Highly Effective
3. Helps students learn at a rate that is comfortable to them, and that allows them to retain information.	4.34	Highly Effective	4.68	Highly Effective
4. Facilitates comprehension of abstract concept and life skills.	4.23	Effective	4.52	Highly Effective
5. Enables students' to express their ideas and thoughts better.	4.37	Highly Effective	4.68	Highly Effective
6. Promotes active and engaging lesson for students' best learning experience.	4.31	Highly Effective	4.68	Highly Effective
7. Enables the students' to be more active and engaged in the lesson.	4.26	Highly Effective	4.72	Highly Effective
8. Improves motivation and engagement in the learning process.	4.34	Highly Effective	4.68	Highly Effective
9. Provides different opportunities to make learning more fun and enjoyable in terms of teaching same things in new ways.	4.37	Highly Effective	4.72	Highly Effective
10. Encourages a more active participation in the learning process which can be hard to achieve through a traditional lecture environment.	4.54	Highly Effective	4.84	Highly Effective
Over-all Mean	4.35	Highly Effective	4.70	Highly Effective

Extent of Effectiveness in the Utilization of Information and Communication Technology in Terms of Delivery of Lesson According to Average Family Monthly Income

Table 13 shows the extent of effectiveness of the utilization of information and communication technology in terms of delivery of lesson according to average family monthly income. The lower income group obtained the highest mean score of 4.47 on Item No. 10, "Encourages a more active participation in the learning process" which can be hard to achieved through a traditional lecture environment. The lowest mean score achieve was 4.20, interpreted "Highly Effective" as on Item No. 4, "Facilitates comprehension of abstract concept and life skills". The

Psych Educ, 2023, **14**: 715-734, Document ID:2023 PEMJ1289, doi:10.5281/zenodo.10032222, ISSN 2822-4353
Research Article

overall mean score was 4.31, interpreted as "Highly Effective" which implies that the teachers with lower income experienced that ICT was Highly Effective in the delivery of lessons.

The group of teachers with higher income have common perception on the extent of effectiveness of the utilization of information and communication technology in terms of delivery of lesson. They obtained the highest mean score of 4.87 in Item No. 10, "Encourages a more active participation in the learning process" which can be hard to achieve through a traditional lecture environment. Conversely, they obtained the lowest mean score of 4.50, interpreted as "Highly Effective", which was evident on Item No. 4, Facilitates comprehension of abstract concept and life skills. The over-all mean was 4.68, interpreted as "Highly Effective". The result indicates that the extent of effectiveness in the utilization of information and communication technology in terms of delivery of lesson was "High".

Dermont (2016) added that computer tools help students and teachers manipulate complex data-sets. This then provides the context for effective discussion that help to develop subject understanding. ICT is beneficial for teachers to share resources, expertise and advice. It is also easier to plan and prepare lessons and design materials for students. Sometimes, ICT helps teachers to access up-to-date students and school data, anytime anywhere. Teachers can enhance their professional image by using ICT.

Table 13. *Extent of Effectiveness of the Utilization of Information and Communication Technology in Terms of Delivery of Lesson According to Average Family Monthly Income*

	Lower		*Higher*	
Items	*Mean*	*Interpretation*	*Mean*	*Interpretation*
1. Improves the quality of teaching.	4.40	Highly Effective	4.73	Highly Effective
2. Makes delivery of lesson more interactive.	4.27	Highly Effective	4.70	Highly Effective
3. Helps students learn at a rate that is comfortable to them, and that allows them to retain information.	4.27	Highly Effective	4.70	Highly Effective
4. Facilitates comprehension of abstract concept and life skills.	4.20	Effective	4.50	Highly Effective
5. Enables students' to express their ideas and thoughts better.	4.30	Highly Effective	4.70	Highly Effective
6. Promotes active and engaging lesson for students' best learning experience.	4.37	Highly Effective	4.57	Highly Effective
7. Enables the students' to be more active and engaged in the lesson.	4.27	Highly Effective	4.63	Highly Effective
8. Improves motivation and engagement in the learning process.	4.30	Highly Effective	4.67	Highly Effective
9. Provides different opportunities to make learning more fun and enjoyable in terms of teaching same things in new ways.	4.30	Highly Effective	4.73	Highly Effective
10. Encourages a more active participation in the learning process which can be hard to achieve through a traditional lecture environment.	4.47	Highly Effective	4.87	Highly Effective
Over-all Mean	4.31	Highly Effective	4.68	Highly Effective

Extent of Effectiveness of the Utilization of Information and Communication Technology in Terms of Monitoring and Assessment According to Age

Table 14 shows the extent of effectiveness of the utilization of information and communication technology in terms of monitoring and assessment according to age. When younger teachers assessed the effectiveness of the utilization of information and communication technology in terms of monitoring and assessment, they achieve a highest mean score of 4.89, interpreted as "Highly Effective" on Item No. 1, "Improves the quality of teaching". On the other hand, the lowest mean score of 4.63 was obtained on Item No. 6, "Promotes active and engaging lesson for students' best learning experience". The over-all mean score was 4.77, interpreted as "Highly Effective".

The result denotes that the younger age group of teachers observe that the ICT is highly effective in monitoring and assessment. Broadly speaking, schools recognize that systems can improve effectiveness and reduce costs. Across schools, using ICT to manage data was found to promote teaching and learning by facilitating more effective timetabling. The most effective tools were found to be school- devised systems and the use of Excel spreadsheets. As we know, school data was in forming the setting and in compiling reports to parents (Lucas, 2015).

Psych Educ, 2023, **14**: 715-734, Document ID:2023 PEMJ1289, doi:10.5281/zenodo.10032222, ISSN 2822-4353
Research Article

Table 14. *Extent of Effectiveness of the Utilization of Information and Communication Technology in Terms of Monitoring and Assessment According to Age*

Items	Younger Mean	Younger Interpretation	Older Mean	Older Interpretation
1. Improves the quality of teaching.	4.89	Highly Effective	4.48	Highly Effective
2. Makes delivery of lesson more interactive.	4.85	Highly Effective	4.48	Highly Effective
3. Helps students learn at a rate that is comfortable to them, and that allows them to retain information.	4.85	Highly Effective	4.48	Highly Effective
4. Facilitates comprehension of abstract concept and life skills.	4.78	Highly Effective	4.52	Highly Effective
5. Enables students' to express their ideas and thoughts better.	4.70	Highly Effective	4.45	Highly Effective
6. Promotes active and engaging lesson for students' best learning experience.	4.63	Highly Effective	4.45	Highly Effective
7. Enables the students' to be more active and engaged in the lesson.	4.78	Highly Effective	4.48	Highly Effective
8. Improves motivation and engagement in the learning process.	4.74	Highly Effective	4.52	Highly Effective
9. Provides different opportunities to make learning more fun and enjoyable in terms of teaching same things in new ways.	4.81	Highly Effective	4.55	Highly Effective
10. Encourages a more active participation in the learning process which can be hard to achieve through a traditional lecture environment.	4.70	Highly Effective	4.52	Highly Effective
Over-all Mean	4.77	Highly Effective	4.49	Highly Effective

Extent of Effectiveness of the Utilization of Information and Communication Technology in Terms of Monitoring and Assessment According to Length of Service

Table 15 shows the extent of effectiveness of the utilization of information and communication technology in terms of monitoring and assessment according to length of service. Teachers with shorter length of service obtained the highest mean score of 4.85, interpreted as "Highly Effective" on Item No. 1, "Improves the quality of teaching". In the contrary, the lowest mean score of 4.63 was obtained on Item No. 6, Promotes active and engaging lesson for students' best learning experience". The overall mean score was 4.75, interpreted as "Highly Effective, indicating that the teachers with shorter length of service construed that the ICT was "Highly Effective" in monitoring and assessment.

Table 15. *Extent of Effectiveness of the Utilization of Information and Communication Technology in Terms of Monitoring and Assessment According to Length of Service*

Items	Shorter Mean	Shorter Interpretation	Longer Mean	Longer Interpretation
1. Improves the quality of teaching.	4.85	Highly Effective	4.52	Highly Effective
2. Makes delivery of lesson more interactive.	4.81	Highly Effective	4.52	Highly Effective
3. Helps students learn at a rate that is comfortable to them, and that allows them to retain information.	4.81	Highly Effective	4.52	Highly Effective
4. Facilitates comprehension of abstract concept and life skills.	4.78	Highly Effective	4.52	Highly Effective
5. Enables students' to express their ideas and thoughts better.	4.67	Highly Effective	4.48	Highly Effective
6. Promotes active and engaging lesson for students' best learning experience.	4.63	Highly Effective	4.45	Highly Effective
7. Enables the students' to be more active and engaged in the lesson.	4.78	Highly Effective	4.48	Highly Effective
8. Improves motivation and engagement in the learning process.	4.70	Highly Effective	4.55	Highly Effective
9. Provides different opportunities to make learning more fun and enjoyable in terms of teaching same things in new ways.	4.78	Highly Effective	4.58	Highly Effective
10. Encourages a more active participation in the learning process which can be hard to achieve through a traditional lecture environment.	4.67	Highly Effective	4.55	Highly Effective
Over-all Mean	4.75	Highly Effective	4.52	Highly Effective

Teachers with longer length of service obtained the highest mean score of 4.58, interpreted as "Highly Effective" on Item No. 9, "Provides different opportunities to make learning more fun and enjoyable in terms of teaching same things in new ways". Contrary, the lowest mean score of 4.45, interpreted as "Highly Effective" was on Item No. 6, "Promotes active and engaging lesson for students' best learning experience". The over-all mean score was 4.52, interpreted as "Highly Effective", indicating that the extent of effectiveness of the utilization of information and communication technology in terms of monitoring and assessment was "High".

Tao (2016) explained that the quality of technology integration was related to the years of teacher service.

Henry (2012) revealed a positive relationship was identified that indicated that as the years of experience of teachers increased, the level of technology implementation also tended to increase.

Extent of Effectiveness of the Utilization of Information and Communication Technology in Terms of Monitoring and Assessment According to Highest Educational Attainment

Table 16 shows the extent of effectiveness of the utilization of information and communication technology in terms of monitoring and assessment according to highest educational attainment. Teachers with bachelor's degree obtained the highest mean score of 4.5, interpreted as "Highly Effective" on Item 2 and 9, "Makes delivery of lesson more interactive" and "Provides different opportunities to make learning more fun and enjoyable in terms of teaching same things in new ways". On the other hand, the lowest mean score obtained was 4.37, interpreted as "Highly Effective" was on Item No. 5 and 6, 5" Enables students' to express their ideas and thoughts better" and "Promotes active and engaging lesson for students' best learning experience". The over-all mean score was 4.47, interpreted as "Highly Effective", indicating that the teachers with bachelor's degree observed that the ICT was "Highly Effective" in monitoring and assessment.

When teachers with master's degree evaluated the extent of effectiveness of the utilization of information and communication technology in terms of monitoring and assessment. The highest mean score obtained was 4.88, interpreted as "Highly Effective" on 1,3, 9 and 10, "Improves the quality of teaching". "Helps students learn at a rate that is comfortable to them, and that allows them to retain information". "Provides different opportunities to make learning more fun and enjoyable in terms of teaching same things in new ways". "Encourages a more active participation in the learning process which can be hard to achieve through a traditional lecture environment". Conversely, the lowest mean score of 4.76, interpreted as "Highly Effective" was on Item No. 6 and 8, "Promotes active and engaging lesson for students' best learning experience". "Improves motivation and engagement in the learning process".

Overall, the mean was 4.84, interpreted as "Highly Effective". The result implies that the teachers with master's degree justifies that ICT was "Highly Effective".

Professional development is one means that can change the self -efficacy of a teacher and the level that they integrate technology in the classroom. High quality professional development is central to any education improvement effort, especially those that pertain to the integration of technology to support classroom instruction. Professional development is useful in providing teachers with the knowledge and practice they need to implement technology successfully. More specifically, professional development has been identified as one of the most important factors influencing teachers' integration of technology into the classroom (Pellegrino, 2012).

Table 16. *Extent of Effectiveness of the Utilization of Information and Communication Technology in Terms of Monitoring and Assessment According to Highest Educational Attainment*

	Bachelor		*Masters*	
Items	*Mean*	*Interpretation*	*Mean*	*Interpretation*
1. Improves the quality of teaching.	4.51	Highly Effective	4.88	Highly Effective
2. Makes delivery of lesson more interactive.	4.51	Highly Effective	4.84	Highly Effective
3. Helps students learn at a rate that is comfortable to them, and that allows them to retain information.	4.49	Highly Effective	4.88	Highly Effective
4. Facilitates comprehension of abstract concept and life skills.	4.49	Highly Effective	4.84	Highly Effective
5. Enables students' to express their ideas and thoughts better.	4.37	Highly Effective	4.84	Highly Effective
6. Promotes active and engaging lesson for students' best learning experience.	4.37	Highly Effective	4.76	Highly Effective
7. Enables the students' to be more active and engaged in the lesson.	4.49	Highly Effective	4.80	Highly Effective
8. Improves motivation and engagement in the learning process.	4.51	Highly Effective	4.76	Highly Effective
9. Provides different opportunities to make learning more fun and enjoyable in terms of teaching same things in new ways.	4.51	Highly Effective	4.88	Highly Effective
10. Encourages a more active participation in the learning process which can be hard to achieve through a traditional lecture environment.	4.40	Highly Effective	4.88	Highly Effective
Over-all Mean	4.47	Highly Effective	4.84	Highly Effective

Successful implementation of educational technologies depends on high quality professional development

Psych Educ, 2023, **14**: 715-734, Document ID:2023 PEMJ1289, doi:10.5281/zenodo.10032222, ISSN 2822-4353
Research Article

along with ongoing support. Teachers who have successfully integrated technology in their classrooms have reported participating in professional development that helped them understand how curriculum, standards, and technologies connect (Penuel, 2012).

Extent of Effectiveness of the Utilization of Information and Communication Technology in Terms of Monitoring and Assessment According to Average Family Monthly Income

Table 17. *Extent of Effectiveness of the Utilization of Information and Communication Technology in Terms of Monitoring and Assessment According to Average Family Monthly Income*

Items	Lower		Higher	
	Mean	Interpretation	Mean	Interpretation
1. Improves the quality of teaching.	4.47	Highly Effective	4.87	Highly Effective
2. Makes delivery of lesson more interactive.	4.50	Highly Effective	4.80	Highly Effective
3. Helps students learn at a rate that is comfortable to them, and that allows them to retain information.	4.47	Highly Effective	4.83	Highly Effective
4. Facilitates comprehension of abstract concept and life skills.	4.43	Highly Effective	4.83	Highly Effective
5. Enables students' to express their ideas and thoughts better.	4.30	Highly Effective	4.83	Highly Effective
6. Promotes active and engaging lesson for students' best learning experience.	4.20	Effective	4.87	Highly Effective
7. Enables the students' to be more active and engaged in the lesson.	4.33	Highly Effective	4.90	Highly Effective
8. Improves motivation and engagement in the learning process.	4.43	Highly Effective	4.80	Highly Effective
9. Provides different opportunities to make learning more fun and enjoyable in terms of teaching same things in new ways.	4.43	Highly Effective	4.90	Highly Effective
10. Encourages a more active participation in the learning process which can be hard to achieve through a traditional lecture environment.	4.33	Highly Effective	4.87	Highly Effective
Over-all Mean	4.39	Highly Effective	4.85	Highly Effective

Table 17 presents the extent of effectiveness of the utilization of information and communication technology in terms of monitoring and assessment according to average family monthly income. Teachers who had lower family monthly income garnered a highest mean score of 4.50, interpreted as "Highly Effective" in Items No. 2, "Makes delivery of lesson more interactive, interpreted as highly effective". In the contrary, the lowest mean score obtained was 4.20, interpreted as "Highly Effective". Overall the mean score was 4.39, interpreted as "Highly Effective". The result denotes that the with lower family monthly income believes that the utilization of ICT was "Highly Effective".

When teachers with higher family monthly income evaluated the effectiveness of ICT utilization, they garnered a highest mean score of 4.90, interpreted as "Highly Effective" on Item No. 7 and 9. "Enables the students' to be more active and engaged in the lesson", "Provides different opportunities to make learning more fun and enjoyable in terms of teaching same things in new ways". The over-all mean score was 4.85, interpreted as "Highly Effective".

The result indicates that teachers with higher monthly family income agreed that the use of ICT was highly effective in the delivery of learning from the pupils. ICT may contribute to creating powerful learning environments in numerous ways. The data suggest that the majority of teachers valued the potential contribution of ICT and had initially held not only positive attitudes towards but great enthusiasm for ICT use teaching (Smeets, 2014). Leem (2015) found that teachers are using ICT to change their role from that of primary source of information to one who provides students with structure and advice, monitors their progress and assesses their accomplishments.

Academic Performance of Pupils

The academic performance of the schools in Binalbagan District II was "satisfactory". The result can be attributed to the effectiveness in the utilization of Information and Communication Technology, which helps teachers in planning the lesson, delivery of the lesson and monitoring and evaluation.

Table 18. *Academic Performance of Pupils*

School	*Mean*	*Interpretation*
School A	80.97	Satisfactory
School B	83.02	Satisfactory
School C	82.33	Satisfactory
School D	85.53	Very Satisfactory
School E	84.83	Satisfactory
School F	84.77	Satisfactory
School G	85.43	Very Satisfactory
School H	85.18	Very Satisfactory
School I	85.41	Very Satisfactory
School J	85.00	Very Satisfactory
School K	85.73	Very Satisfactory
School L	85.33	Very Satisfactory
School M	87.23	Very Satisfactory
School N	87.42	Very Satisfactory
Over all Mean	84.87	Satisfactory

Difference in the Extent of Effectiveness of the Teachers' Utilization of Information and Communication Technology in Terms of Planning the Lesson According to Variables

Mann Whitney U test was utilized to test the difference in the extent of effectiveness of the teachers' utilization of information and communication technology in terms of planning the lesson according to variables. In terms of age, it obtained a *p* value of 0.117, length of service with the *p* value of 0.300, highest educational attainment with the *p* value of 0.003 and average family monthly income with the *p* value of 0.001.

The result indicated that there was no significant difference in the extent of effectiveness of the teachers' utilization of information and communication technology in terms of planning the lesson according to age and length of service. The null hypothesis is, therefore, accepted. Conversely, a significant difference was found on the extent of effectiveness of the teachers' utilization of information and communication in terms of planning the lesson according highest educational attainment and average family monthly income. The null hypothesis which states there is no significant difference in the extent f effectiveness of the teachers' utilization of ICT when grouped and compared according to the aforementioned variables, in this regard was rejected.

The utilization of information and communication technologies can help revitalize teachers and students. This can help to improve and develop the quality of education by providing curricular support in difficult subject areas. Educational attainment of the teachers reflects the effectiveness of their utilization of the information and communication technologies (Sanguyo, 2016).

Further, Goza (2015) added that family income of teachers may influence the utilization of educational technology. Their capacity to acquire such technology may be relative to its utilization.

Table 19. *Difference in the Extent of Effectiveness of the Teachers' Utilization of Information and Communication Technology in Terms of Planning of Lesson According to Variables*

Variable	*Category*	*Mean*	*Z*	*p-value*	*Sig level*	*Interpretation*
Age	Younger	4.56	1.476	0.145	0.05	Not Significant
	Older	4.39				
Length of Service	Shorter	4.56	1.354	0.181		Not Significant
	Longer	4.40				
Highest Educational Attainment	Bachelor	4.29	-4.059	0.000		Significant
	Masters	4.72				
Average Family Monthly Income	Lower	4.21	-5.538	0.000		Significant
	Higher	4.73				

Difference in the Extent of Effectiveness of the Teachers' Utilization of Information and Communication Technology in Terms of Delivery of Lesson According to Variables

Table 20 shows the difference in the extent of effectiveness of the teachers' utilization of information and communication technology in terms of delivery of lesson according to variables. In terms of age, it garnered a *p* value of 0.017 which was higher than the 0.05 level of significance. Correspondingly, as to length of service, the *p* value was 0.049 which was likewise greater than the 0.05 level of significance indicating that the difference was not significant. The null hypothesis in this regard was therefore accepted.

As to highest educational attainment and average

Psych Educ, 2023, **14**: 715-734, Document ID:2023 PEMJ1289, doi:10.5281/zenodo.10032222, ISSN 2822-4353
Research Article

family monthly income, the *p* value was 0.003 and 0.001 which was lower than the 0.05 level of significance denoting that the difference was significant. The null hypothesis was therefore rejected.

Teachers have a central role in providing learning by making use of available techniques in the delivery of lessons towards students. Sorento (2016) argue that the educational attainment of the teachers contributed to their level of competence in the utilization of ICT. Further, Golam(2015), in his study concluded that family income is one of the factors that affect teachers on their extent of utilization of information and communication technology.

Table 20. *Difference in the Extent of Effectiveness of the Teachers' Utilization of Information and Communication Technology in terms of Delivery of Lesson According to Variables*

Variable	*Category*	*Mean*	*Z*	*p-value*	*Sig level*	*Interpretation*
Age	Younger	4.60	1.592	0.117	0.05	Not Significant
	Older	4.42				
Length of Service	Shorter	4.56	1.047	0.300		Not Significant
	Longer	4.44				
Highest Educational Attainment	Bachelor	4.35	-3.153	0.003		Significant
	Masters	4.70				
Average Family Monthly Income	Lower	4.31	-3.486	0.001		Significant
	Higher	4.68				

Difference in the Extent of Effectiveness of the Teachers' Utilization of Information and Communication Technology in Terms of Monitoring and Assessment According to Variables

Table 21 shows the difference in the extent of effectiveness of the teachers' utilization of information and communication technology in terms of monitoring and assessment according to variables. Using the Mann Whitney U test the p values obtained in according to profile variables was 0.017, 0.049, 0.002 and 0.000, which was lesser than the 0.05 level of significance indicating the difference was "Significant".

The result implies that there was a significant difference in the extent of effectiveness of the teachers' utilization of information and communication technology in terms of monitoring and assessment according to variables. Further it indicates that the profile of the teachers influence the extent of effectiveness of the utilization of information and communication technology in terms of monitoring and assessment according to variables.

The use of almost all types and forms of educational information resources in educational practice substantially improves the quality of monitoring and assessment among students. It makes information accessible, updated and more dynamic (Dizon, 2016).

Table 21. *Difference in the Extent of Effectiveness of the Teachers' Utilization of Information and Communication in Terms of Monitoring and Assessment According to Variables*

Variable	*Category*	*Mean*	*Z*	*p-value*	*Sig level*	*Interpretation*
Age	Younger	4.77	2.447	0.017	0.05	Significant
	Older	4.49				
Length of Service	Shorter	4.75	2.003	0.049		Significant
	Longer	4.52				
Highest Educational Attainment	Bachelor	4.47	-3.330	0.002		Significant
	Masters	4.84				
Average Family Monthly Income	Lower	4.39	-4.453	0.000		Significant
	Higher	4.85				

Relationship Between the Extent of Effectiveness of the Utilization of Information and Communication Technology and Pupils' Academic Performance

Table 22 shows the relationship between the extent of effectiveness of the utilization of information and communication technology and pupils' academic performance. The *p* value obtained was 0.460 which indicated that the relationship was "Not Significant". The result implies that the effectiveness of the utilization of information and communication technology is not relative to pupils' academic performance.

According to the results, technology usage was indirectly correlative and statistically significant with pupils' academic performance. The findings is in agreement with other studies of Taradi and his colleagues (2015) who found that the use of technology affected learning outcomes to a minimal level. This finding is very motivating, especially at the University of Dammam. Furthermore, research has indicated that a comparatively higher degree of students learning and effective teaching can be achieved without the use of technology.

Furthermore, research has indicated that the use of technology can support new instructional approaches and make hard-to-implement instructional methods such as simulation or cooperative learning more feasible. Moreover, educators commonly agree that technology has the potential to improve student learning outcomes and effectiveness. Integration has a sense of completeness or wholeness, by which all essential elements of a system are seamlessly combined together to make a whole, yet its effectiveness may not influence student's academic outcomes (Chang & Wu, 2016).

Table 22. *Relationship Between the Extent of Effectiveness of the Utilization of Information and Communication Technology and Pupils' Academic Performance*

	r	*p-value*	*Interpretation*
Extent of Effectiveness of the Utilization of ICT Pupil's Academic Performance	0.097	0.460	Not significant

Problems Encountered By The Teachers In The Utilization Of Information And Communication Technology

Table 23 shows the problems encountered by the teachers in the utilization of information and communication technology. The problems were ranked according to its degree. Item No. 1, Lack or inadequate ICT equipment was ranked as first. Item No. 5, Not willing to learn ICT was ranked as the last among the problems presented.

Based on the findings of the study it was found out that there was lack or inadequate ICT equipment used by the teachers. Some of them have their personal computers and use if for school works. Although the resources was inadequate teachers who were able to utilize it was able to prove that it was highly effective.

Table 23. *Problems Encountered by the Teachers in the Utilization of Information and Communication Technology*

Problems	*F*	*Rank*
1. Lack or inadequate ICT equipment.	58	1
2. Not familiar with latest applications and software like PPT, Word and Excel.	23	6.5
3. Lack of effective pedagogical training .	49	2
4. Need to use personal money to procure ICT equipment.	42	3
5. Not willing to learn ICT.	14	10
6. Lacked basic ICT skills.	23	6.5
7. Limited numbers of internet connections.	33	4
8. Lack of funds to purchase or upgrade hardware and/or software.	26	5
9. Time limitations, and pressure to cover the curriculum.	21	8
10. Cannot afford technicians to support educational computing.	16	9

Conclusion

(1) Older teachers, who had longer years in service and had bachelor's degree who belong to lower and higher income groups participated in this study. (2) Teachers believed that the utilization of ICT is highly effective in planning the lesson, delivery of the lesson and monitoring and evaluation. (3) There was a high level of effectiveness in the utilization of ICT by the teachers according to profile. (4) Pupils obtained a satisfactory academic performance. (5) Teachers' age and length of service do not influence their perception on the extent of effectiveness of utilization of ICT in planning the lesson and delivery of lesson. (6) The extent of effectiveness of the utilization of information and communication technology does not affect pupils' academic performance.

Based on the findings and conclusions of this study, the following recommendations was used.

The Department of Education should provide sufficient resources for teachers to be able to incorporate technology use into the curriculum and to develop technology-centered lessons to increase student-centered learning requiring technology applications.

School principals of the District of Binalbagan II need to support and encourage their teachers to use the available technology and to participate in professional development technology training provided outside of what the school and district offers. Continuous professional development will increase teachers' knowledge and experience of technology, and reduce teachers' technology anxiety. They need to continue to encourage technology integration. Continued funding for technology equipment and professional learning that focuses on instructional application of technology.

Teachers of the District of Binalbagan II shall continue to upgrade their selves with the current issues on the utilization of ICT. They shall attend seminars and skills training to enhance the effectiveness of the utilization of the ICT.

References

Castro, K.P. (2016). In-service teachers' attitudes towards the use of information and communication technology in teaching practice: the case of Jordan. Conference IMETC 2016.

Chang, C.E. & Wu,R.O.(2016). An exploration of Jordanian teachers' attitudes, skills, and access as indicator of ICT integration in Jordan. Amman Jordan.

Cordero, D.E. (2016). Information technology for schools, creating practical knowledge to improve students' performance. Jossey-Bass A Wiley Company San Francisco.

De Belen, R. D. (2015). Educational Research: An Introduction (4th ed.). Philippines.

Dermont, R.E.(2016). The attitudes toward adopting information technology by vocational and technological teachers in southern Taiwan (China). (Doctoral Dissertation, Idaho State University, 2016). ProQuest Digital Dissertations (UMI No. AAT 3082993).

Eglisias, M.P. (2015). The IEA study of computers in education: Implementation of an innovation in 21 education systems, Pergamon Press.

Froelan, D.G. (2015). Teaching and learning for the future. Report of the Committee on Multi Media in Teacher Training. Den Haag: SDU.

Gonzalo, T.L.(2016). Teacher and teacher-directed student use of computers and software. Center for Research on Information Technology and Organizations, University of California, Irvine, and University of Minnesota.

Gorder, M. D. (2015). Bellingham Public Schools Technology and Learning Implementation Plan.

Henry, P. S. (2012). Computers in the classroom: What shall I do? New York: Garland Publishing, Inc.

Hermans, A.S. (2016). Factors influencing the effective use of technology for teaching and learning: lesson learned from the SEIR-TEC intensive site schools. Southeast and Islands Regional Technology in Education Consortium.

Homes, M.L. (2013). Information & Communication Technologies (ICT) in Education for Development.

Honey, B.N. (2015). The power of digital learning: Integrating digital content. CEO Forum School Technology and Readiness. Year 3 Report.

Hubona. K.O. (2015). Federal support for technology in k-12 education. Brookings Paper on Educational Technology.

Inan, F.T. (2015). America's technology literacy challenge – Overview. [On-line].The White House, Available: http://clinton4.nara.gov/WH/New/edtech/0cont.html

Kim, W.V. (2015). Teachers and machines: the classroom use of technology since 1920. New York: Teachers College Press.

Koh , N.H.(2015). Oversold & underused: computers in the classroom. Cambridge: The Harvard University Press.

Leem, N.L. (2015). Teaching faculty how to use technology: Best practices from leading institutions. Connecticut: The Oryx Press.

Mabolos, V.N.(2015). Principals, technology, and change. The Technology Source. Retrieved May 17, 2015.

Martin, F.H. (2015). Evaluation and selection of web course management tools. Retrieved January 13, 2018, fromhttp:sunil.umd.edu/webct.

McAdoo, F.L.(2015). Technology in it place: successful technology infusion in schools. San Franscisco: Jossey-Bass Inc.

McConnell, W.T.(2016). Computers for twenty-first century educators. New Jersey: Pearson Education, Inc.

Mueller, H.P. (2014). Windows on the future: Education in the age of technology. California: Corwin Press, Inc.

Okauru, S.E.(2011). Educational Leadership: Principals as Leaders in a Culture of Change.

Oughden,P.O.(2012). Foundation for Information Technology for Education and Development.

Pelbert, D.D.(2016). Computers in education. Boston: Allyn and Bacon.

Pellegrino, F.L.(2012). The biography of a printing press. Berkeley: University of California Press.

Penuela, T.R.(2016). Computers in the schools: A guide to planning and administration. Toronto: Macmillan Publishing Company.

Poole. L.P. (2013). Technically speaking: Why all Americans need to know more about technology. Washington, DC: National Academy Press.

Redmann, B.T. (2015). Trends and issues in instructional design and technology. Columbus: Merrill Prentice Hall.

Roca, R.L. (2015). The impact of media and technology in schools: A research report prepared for The Bertelsmann Foundation.

Rowand, K.O. (2015). The learning return on our educational technology investment. San Francisco: WestEd. Retrieved from: http://www.wested.org/cs/we /view/rs/619.

Sang, S.S.(2014). Integrating educational technology into teaching. New Jersey: Pearson Education, Inc.

Sanguyo, A.L.(2016). Diffusion of innovations (4th ed.). New York: Free Press.

Sasuman, R.O. (2015). Integrating computer use in primary schools: The challenge for principals. Paper presented at the Australian Council for Educational Administration Annual Conference, Hobart, Australia.

Scheffler, L. B. (2015). Empowering teachers with technology: Making it Happen. Maryland: The Scarecrow Press, Inc.

Simmons, R.I.(2014). Enabling Learning at the Workplace – Humans first, Technology second: Adopting a Best-Fit Approach to Implementing ICT in Training.

Sorento, M.O. (2016). Teachers and technology: Making the connection. (Report No. OTA-HER-616). Washington, DC: Government Printing Office.

Soyan, A.L.(2014). Toward a new golden age in American education: How the internet, the law and today's students are revolutionizing expectations.

Taradi, S.L.(2015) Teaching and learning with information and communication technologies (ICT) for numeracy in the early childhood and primary years of schooling. Australia: Department of Education, Training and Youth Affairs

Torendon, U.P. (2015). ICT and multimedia in the primary school. Paper presented at the 16th conference on educational uses of information and communication technologies, Beijing, China

Vermette, R.O.(2015). The impact of information and communication technologies on learning and attainment. Full report, March 2003. http://camara.

Wang, D.S.(2014). What factors support or prevent teachers from using ICT in their classrooms. Paper presented at the British Educational Research Association Annual Conference.

Weden, A.A. (2014). Barriers to the integration of the Internet into teaching and learning: Professional development. Paper presented at the Asia Pacific Regional Internet Conference on Operational

Affiliations and Corresponding Information

Athena Lourdes S. Esmaquilan
- STI West Negros University – Philippines
- Payao Elementary School
  Department of Education – Philippines

Josyl S. Edaño
- STI West Negros University – Philippines
- Payao Elementary School
  Department of Education – Philippines

Josielee Grace S. Vasquez
- STI West Negros University – Philippines
- Wes Visayas University – Philippines